Relation-Level Structural Analysis of Predictive Multiplicity in Knowledge Graph Embedding for Link Prediction

知識グラフ埋め込みにおける予測多重性をもたらす
関係レベル構造要因の解明

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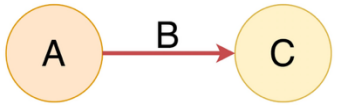
Background: Knowledge graph embedding link prediction

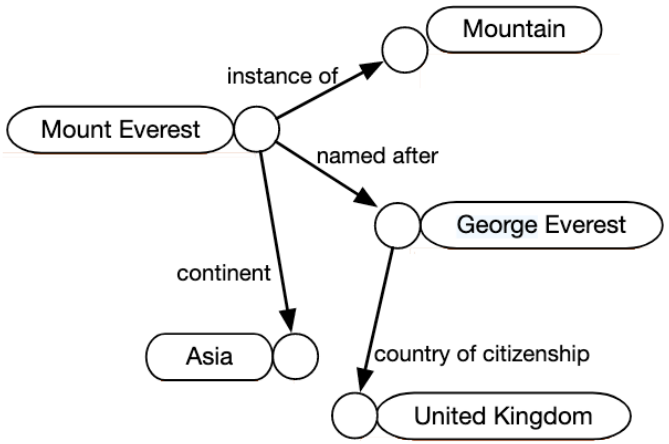


Triple

A :head entity

B :relation

C :tail entity



Example of Knowledge graph[1]

Knowledge graph link prediction

(Tail prediction & Head prediction)

Goal: $(h, r, ?)$ or $(?, r, t)$

(Paris, capital_of , ?)

Predicted ranking (2 circumstances)

Ranking	Entity
1	France
2	Germany
3	Italy
...	...
10	Spain
11	China
...	...

$Hit@10$

Ranking	Entity
1	Germany
2	Spain
3	Italy
...	...
10	China
11	France
...	...

$Miss@10$

Evaluation metric: $Hits@10$

The proportion of test queries for which this top-10 success condition is satisfied.

Prior Work: Predictive Multiplicity

Predictive multiplicity[2]

Even when models have similar overall performance, they may still disagree on whether a specific query is a top-K hit or miss.

By quantified by **ambiguity** and **discrepancy**.

Ambiguity: among all test queries, how many receive **at least one** conflicting decision?
Discrepancy: for one competing model, how large can the **overall disagreement rate** be?

Example

Test Triples (optimal decision)

```
<CompanyA, has_supplier, SupplierA>
<CompanyB, has_supplier, SupplierB>
```

Query for Model 1, 2:

```
<CompanyA, has_supplier, ?>
<CompanyB, has_supplier, ?>
```

Model 1 Prediction (50% correct)

```
<CompanyA, has_supplier, SupplierA>
<CompanyB, has_supplier, SupplierA>
```

Model 2 Prediction (50% correct)

```
<CompanyA, has_supplier, SupplierB>
<CompanyB, has_supplier, SupplierB>
```

Top-K hit / miss decision for one query-answer pair

$$T_K(M_\theta, \mathrm{tr}(q,e)) = \mathbf{1}\left[R_{\succeq_{M_\theta,q}}(e) \le K\right]$$

Query-level disagreement

$$\Delta(M_\theta, \tau) = \mathbf{1}[T_K(M_\theta, \mathrm{tr}(\tau)) \neq T_K(M_\theta^*, \mathrm{tr}(\tau))]$$

Ambiguity

$$\alpha_\varepsilon(M_\theta^*) = \frac{1}{|T|}\sum_{\tau\in T}\max_{M_\theta \in S_\varepsilon(M_\theta^*)} \Delta(M_\theta, \tau)$$

Discrepancy

$$\delta_\varepsilon(M_\theta^*) = \max_{M_\theta \in S_\varepsilon(M_\theta^*)} \frac{1}{|T|}\sum_{\tau\in T}\Delta(M_\theta, \tau)$$

q : query
e : correct answer entity
$\tau = (q, e)$: one test query-answer pair
$\mathrm{tr}(q, e)$: corresponding triple
M_θ^* : baseline model
M_θ : competing model
T : test set
$S_\varepsilon(M_\theta^*)$: similarly performing models
$R_{\succeq_{M_\theta,q}}(e)$: rank of e for q

Research Focus: From Prior Work to Thesis Question

Original paper

- Formalizes predictive multiplicity in KGE link prediction
- Introduces ambiguity and discrepancy as evaluation metrics
- Shows that multiplicity exists empirically, and voting can mitigate it

My thesis question

- The original paper establishes the existence of predictive multiplicity in KGE link prediction.
- My thesis extends this line of inquiry by asking whether predictive multiplicity is systematically associated with relation-level structure.
- Accordingly, the focus shifts from a global phenomenon to relation-level analysis.
- **Main question: Which relation-level factors are associated with stronger predictive multiplicity?**

Method: Thesis Setting and Relation-Level Analysis Pipeline

Thesis setting

- Dataset: FB15k-237
- Models: RotatE[3] and TransE[4]
- Repeated runs of the same model family
- Analysis unit: relation-level rather than only dataset-level/global-level

FB15k-237: a standard KGE benchmark with less inverse-relation leakage than FB15k, making it more suitable for structural analysis of link prediction.

RotatE / TransE: two representative KGE models used here as a focused cross-model comparison under the current thesis scope.

Pipeline

- Run repeated training with different random seeds
- Compute query-level top-10 hit / miss outcomes
- Aggregate them into relation-level metrics
- Attach relation-level factors
- Compare which factors are associated with stronger multiplicity

Method: Candidate Relation-Level Factors

Mapping type

How many distinct heads/tails are typically associated with a relation.

Typical categories

$1-1$, $1-N$, $M-1$, $M-N$

Intuition

A plausible candidate because reverse structural support may constrain prediction behavior.

Example

- `capital_of`: closer to 1-1
- `born_in`: closer to M-1
- `language_spoken_in`: can look more M-N

Inverse-like support

Whether a relation has a clear reverse partner.

Intuition

If a relation is strongly supported by a reverse relation, repeated runs may learn more consistent behavior

Example

- `parent_of` ⇔ `child_of`
- `capital_of` ⇔ `has_capital`

Symmetry

Whether (h, r, t) is often supported by (t, r, h) under the same relation.

Intuition

Another plausible candidate because internal bidirectional support may affect uncertainty.

Example

- `married_to`
- `adjacent_to`

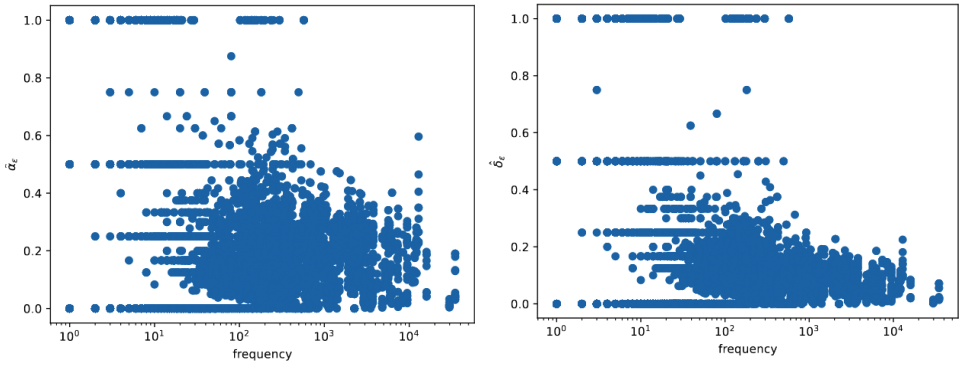
Relation frequency (Support/ sparsity factor)

How often a relation appears in the training graph.

Included as a support/sparsity control variable.

Intuition

Partly motivated by the weak negative relation reported in the original paper.

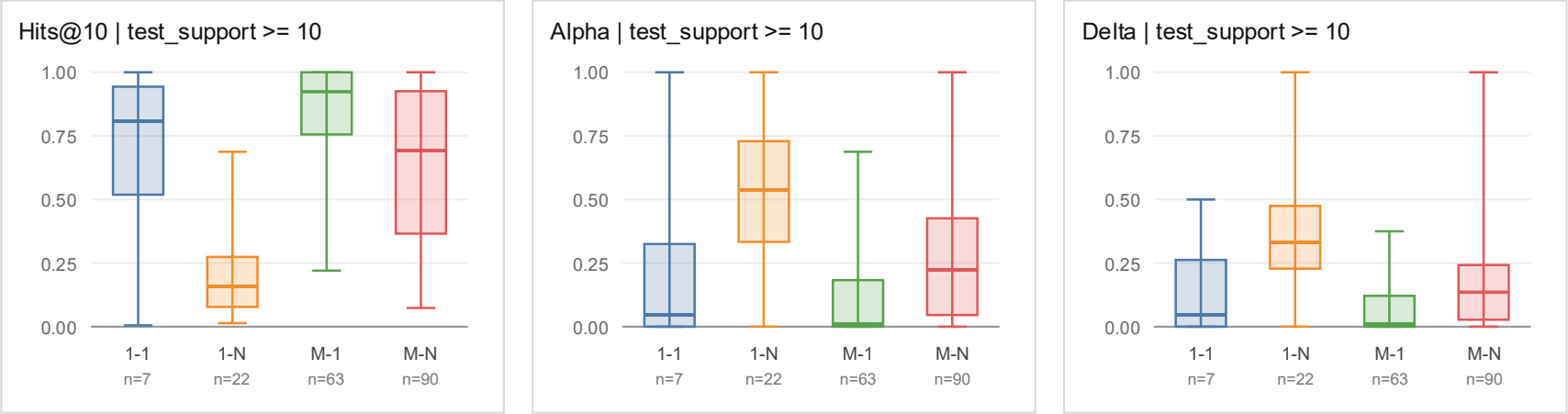


These figures[2] demonstrates the weak negative correlation between relation frequency and empirical ambiguity/ discrepancy.

Main Result: Mapping Type Strongly Affects Multiplicity on Tail Prediction

RotatE tail-side

Query: $(h, r, ?)$

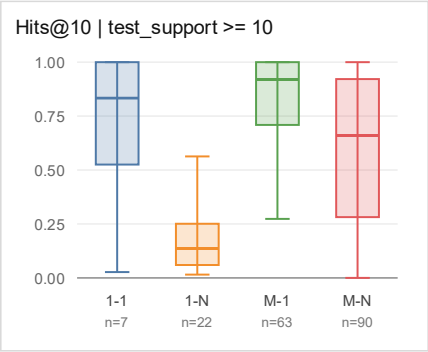


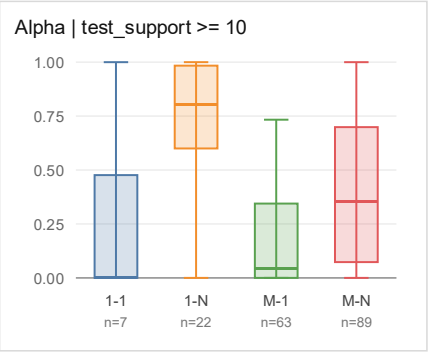
Only relations with test_support >= 10 are shown.

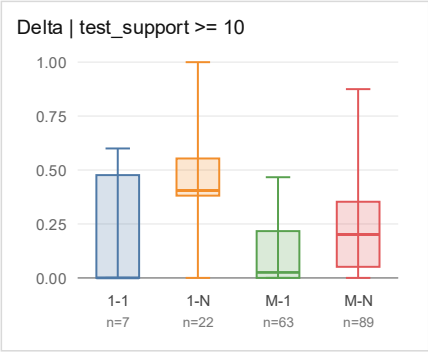
- $1-N$ shows the lowest tail-side $Hits@10$ and the highest multiplicity
- $M-1$ shows the highest tail-side $Hits@10$ and the lowest multiplicity.
- Tail-side predictive multiplicity is not uniformly distributed across mapping types
- Higher multiplicity is associated with lower prediction accuracy

Main Result: Mapping Type Strongly Affects Multiplicity on Tail Prediction

Cross-model check: TransE tail-side Query: $(h, r, ?)$

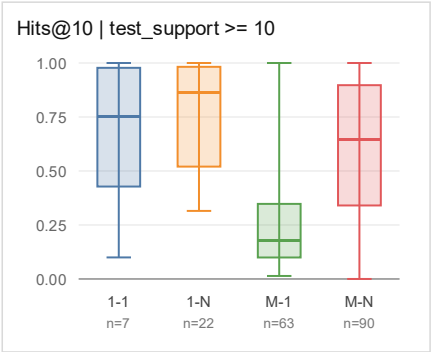


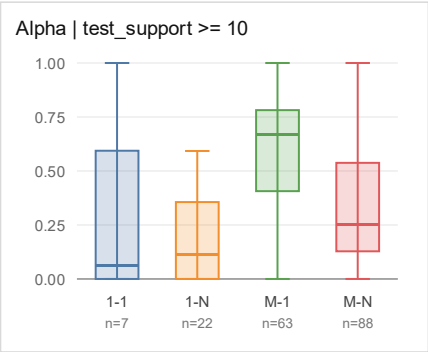


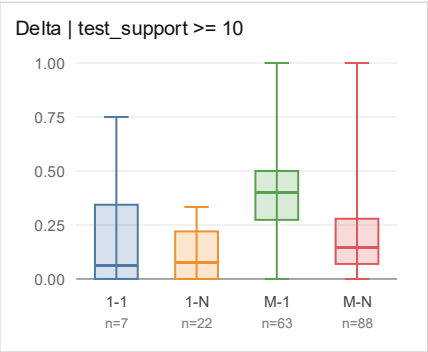


- $M-1$ remains the most stable regime
- $1-N$ remains the most multiplicity-prone regime

By-side check: RotatE head-side Query: $(?, r, \mathrm{t})$







- $1-N$ becomes the most stable regime
- $M-1$ becomes the most multiplicity-prone regime

Main Result: Inverse-Like Support Shows a Small Stable Subgroup

- No clean global monotonic inverse effect is observed.
- A small high-confidence inverse-like subgroup is still visible.
- This subgroup appears in both RotatE and TransE.

High-Confidence Inverse-Like Subgroups (threshold = 10)

Model	Criterion	n	Hits@10	Alpha	Delta
RotatE	High mutual inverse-like strength (mutual_inverse_strength > 0.5)	10	0.6838	0.0971	0.0627
RotatE	High inverse-like clarity (inverse_clarity > 0.5)	5	0.8000	0.0000	0.0000
TransE	High mutual inverse-like strength (mutual_inverse_strength > 0.5)	10	0.6866	0.1361	0.0813

mutual_inverse_strength : Strength. Measuring how strongly a relation and its best reverse partner support each other in both directions.
inverse_clarity: Uniqueness/ confidence. Measuring whether the best inverse-like partner is clearly better than the other candidates.

Example

`/education/educational_institution_campus/educational_institution` → institution of the campus

`/education/educational_institution/campuses` → campuses of the institution

mutual_inverse_strength	=	0.8026
inverse_clarity	=	0.77
hits@10	=	1.0000
alpha	=	0.0000
delta	=	0.0000

Reference

[1] Lv, Yana, and Haomiao Bao. “NSE-KGC: A Knowledge Graph Completion Method Based on Neighborhood Semantic Evidence.” In *2024 5th International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE)*, 287–90, 2024.

[2] Zhu Y, Potyka N, Nayyeri M, et al. Predictive multiplicity of knowledge graph embeddings in link prediction[C]. *Findings of the Association for Computational Linguistics: EMNLP 2024*. 2024: 334-354.

[3] Sun Z, Deng Z H, Nie J Y, et al. Rotate: Knowledge graph embedding by relational rotation in complex space[J]. arXiv preprint arXiv: 1902. 10197, 2019.

[4] Bordes A, Usunier N, Garcia-Duran A, et al. Translating embeddings for modeling multi-relational data[J]. Advances in neural information processing systems, 2013, 26.